

Current Surgical Management of Congenital Diaphragmatic Hernia: A Report From the Congenital Diaphragmatic Hernia Study Group

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Background: Repair of congenital diaphragmatic hernia (CDH) has changed from an emergent procedure to a delayed procedure in the last decade. Many other aspects of management have also evolved since the first successful repair. However, most reports are from single institutions. The lack of a large multicenter database has hampered progress in the management of congenital diaphragmatic hernia (CDH) and makes determination of the current standard difficult.

Methods: The CDH study group was formed in 1995 to collect data from multiple institutions in North America, Europe, and Australia. Participating centers completed a registry form on all live-born infants with CDH during 1995 and 1996. Demographic information, data about surgical management, and outcome were collected for all patients.

Results: Sixty-two centers participated, with 461 patients entered. Overall survival was 280 of 442 patients (63%) where survival was recorded. The defect was left-sided in 78%, right-sided in 21%, and bilateral in 1%. A subcostal approach was used in 91% of patients, with pleural drainage used in 76%. A patch of some kind was used in just over half (51%) of the patients, with polytetrafluoroethylene being the most commonly used material (81%) in those patients with a patch. The mean surgical time was 102 minutes, with an

average blood loss of 14 mL (range, 0 to 500 mL). The overwhelming majority of patients underwent repair between 6:00 AM and 6:00 PM (289 of 329, 88%). Nineteen percent of patients had surgical repair on extracorporeal membrane oxygenation (ECMO) at a mean time of 170 hours into the ECMO course (range, 10 to 593 hours). The mean age at surgery in patients not treated with ECMO was 73 hours (range, 1 to 445 hours).

Conclusions: The multicenter nature of this report makes it a snapshot of current management. The data would indicate that prosthetic patching of the defect has become common, that after-hours repair is infrequent, and that delayed surgical repair has become the preferred approach in many centers. Furthermore, the mean survival rate of 63% indicates that despite decades of individual effort, the CDH problem is far from solved. This highlights the need for a centralized database and cooperative multicenter studies in the future.

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INDEX WORDS: Congenital diaphragmatic hernia, extracorporeal membrane oxygenation, polytetrafluoroethylene, surgical repair.

Congenital diaphragmatic hernia (CDH) in the infant was generally considered a fatal disease through the early 20th century. The report by Ladd and Gross¹ that demonstrated that repair could be undertaken successfully in an infant was a landmark in the surgical management of this disease. They proposed that a policy of waiting for the infant to grow was inadvisable and suggested urgent correction. Gradually this policy evolved so that by 1970, CDH was considered one of the true

surgical emergencies in the newborn.^{2,3} Several reports in the 1980s demonstrated that survival was no different with delayed repair suggesting that a period of stabilization before operation was warranted.⁴⁻⁶ Subsequently there has been a growing acceptance of delayed repair, but the optimal timing of surgery is uncertain. Other controversies in the management of CDH include the use of chest tubes, use of an abdominal versus thoracic approach, and the need for associated procedures.

In 1995, the CDH Study Group (CDHSG) was established as an international effort to collect and analyze multiinstitutional data on this anomaly. One goal of this group was to create a prospective database to establish the current status of disease treatment. This first report describes the current surgical management of CDH based on data gathered from institutions throughout North America, Europe, and Australia.

MATERIALS AND METHODS

An invitation to contribute data to a CDH registry was sent to program directors of all member centers of the Extracorporeal Life Support Organization and all active members of the American Pediatric Surgical Association. An invitation to contribute patients to the registry

The manuscript was prepared by the above-noted investigators for the CDH Study Group; they assume responsibility for the overall contents.

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was also delivered at the Society for Pediatric Research and the Perinatal Section of the American Academy of Pediatrics.

Participants were requested to collect data prospectively on all live-born infants with CDH. The information collected included patient demographics, type of repair, surgical approach, timing of surgery, associated procedures, repair on extracorporeal membrane oxygenation (ECMO), surgical complications, and adjuncts to management such as pleural drainage.

Data collection began from the initial centers in January 1995, and information was collected on each live-born patient with CDH until death or hospital discharge. All registry entries were individually validated with the original data collection form to confirm accuracy of the database input. The data are presented as a percentage of the number of patients with registry fields completed because some fields were empty on some forms. The data were analyzed using an unpaired *t* test or χ^2 analysis where appropriate. Significance was assumed with a *P* value of less than .05.

RESULTS

Data on 461 patients, contributed by 62 centers from North America, Europe, and Australia in 1995 and 1996 (Appendix 1), was available at the time of this report. The overall survival was 280 of 442 (63%) patients where survival was reported. The defect was left sided in 346 (78%), right sided in 92 (21%) and bilateral in five (1%). There was no difference in survival between right- and left-sided defects (*P* = .15). Repair was not performed in 69 patients (15%). Although the reasons that the repair was not performed were not specifically identified, 23 of the 69 patients were on ECMO, eight had chromosomal anomalies, and 20 had cardiac anomalies. The average age of death in the patients not undergoing repair was 2 days in the non-ECMO patients (excluding two patients at 61 and 106 days) and 14 days in the ECMO patients (*P* < .001).

The most common approach for repair was a subcostal incision in 319 (91%) of the patients. A chest tube was inserted in most patients (76%), but a malrotation procedure and appendectomy were infrequently performed (Table 1). Patients with a chest tube had a significantly higher mortality rate than those without a chest tube (29% v 11%, *P* < .01). Some type of patch for repair of the defect was used in about one half of the patients. Quantifying the size of the defect was subjective, but 56 patients were reported to have agenesis of the diaphragm. Of these, 24 (43%) survived. Not surprisingly, patients

Table 1. Associated Procedures

	Yes	No
Incision used		
Subcostal	319 (91%)	—
Thoracic	21 (6%)	—
Other	12 (3%)	—
Chest tube	294 (76%)	91 (24%)
Malrotation procedure	62 (17%)	309 (83%)
Appendectomy	66 (18%)	305 (82%)
Diaphragm patch	234 (51%)	227 (49%)

Table 2. Diaphragm Patch Material Used

Material	No. of Patients (%)
Polytetrafluoroethylene (PTFE)	189 (81)
SILASTIC® (Dow Corning, Midland, MI)	7 (3)
Dacron	4 (2)
Polypropylene (Marlex)	4 (2)
Fascia lata	3 (1)
Muscle flap	2 (1)
Other/unlisted	25 (11)

with a patch repair had a higher mortality rate than patients repaired primarily (44% v 7%, *P* < .001). The most widely used patch material was polytetrafluoroethylene (PTFE), but a range of other materials was also used (Table 2).

The mean surgical time was 102 minutes (range, 18 to 307 minutes). The mean blood loss was 14 mL (range, 0 to 500 mL; median, 5 mL). Significantly more patients had surgical correction performed during "daylight" hours from 6:00 AM to 6:00 PM (*P* < .001, Fig 1). Only eight patients (2%) underwent repair between midnight and 6:00 AM.

ECMO was used in 261 patients (57%), with a survival rate of 54%. The mean duration of ECMO was significantly longer in the nonsurvivors than in the survivors (293 v 228 hours, *P* < .01). Eighty-seven patients had the hernia repaired while on ECMO. The timing of surgery while on bypass ranged from shortly after institution of ECMO to as long as 593 hours into the ECMO course (Fig 2). The timing of repair in patients in whom ECMO was not required was quite variable, ranging from 1 hour after diagnosis to 445 hours (Fig 3). The most common time for repair was within 24 hours (38%), but in many cases surgery was delayed for longer periods.

DISCUSSION

Significant progress in the management of CDH has been hampered by the lack of large series of patients. Most of the published literature on CDH originates from single institutions. Variability in treatment protocols, the

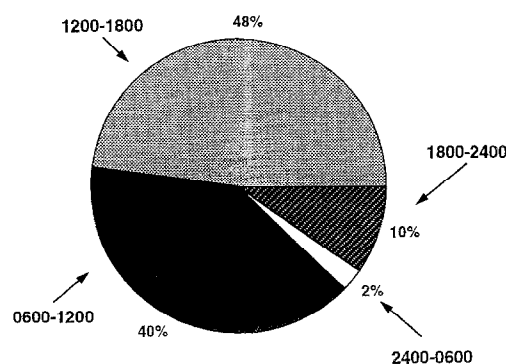


Fig 1. Time of day when the patients underwent repair of the CDH. The number of patients represented is 329.

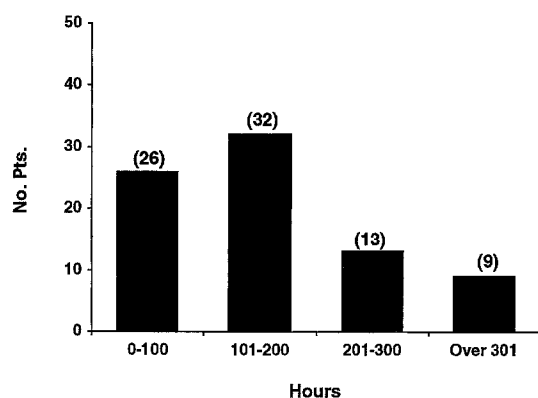


Fig 2. Average age (in hours) of patients undergoing surgery while on ECMO. Numbers in parentheses represent the number of patients in each time period.

limited experience of a single institution, and widespread application of new technologies make it difficult to establish a standard of practice. The CDH study group was organized to gather multiinstitutional data in an effort to improve treatment of these infants. This report represents a summary of the data from a number of institutions and, hence, is a snapshot of the current state of the art. As with any registry-based report, there are some limitations. All data fields for each registry form were not available on every patient. The reports rely on the accuracy of the person entering the data, and there is potential for error in transcribing the data. Regardless, these data do provide the most complete summary to date of the current treatment of infants with CDH.

The earliest reports by Ladd and Gross on correction of CDH in the neonate were encouraging, demonstrating a marked improvement in survival compared with the previous approach of watchful waiting. This prompted most surgeons to correct the defect as early as possible with the hope that removing the bowel and liver from the chest would allow the lungs to expand. Subsequently, Sakai et al,⁶ and others,^{7,8} demonstrated that early opera-

tion actually worsened pulmonary compliance, which sparked a broader interest in delaying operation until the infant was stable. The data presented here clearly show that most pediatric surgeons are currently using a strategy of delay for some period of time before correction of the defect. It remains unclear, however, what constitutes an appropriate period of stabilization.^{9,10} Various investigators have recommended that patients achieve a minimal level of ventilator support, that there be a documented absence of pulmonary hypertension, that there be improvement in pulmonary compliance, or that lung radiographic appearance be improved before repair.^{11,12} As with many aspects of patient management, there is a lack of large controlled trials to identify the most appropriate timing of surgery.¹⁰

The widespread availability of ECMO during the time when delayed repair became more popular led to the use of ECMO to stabilize critically ill infants.¹³⁻¹⁵ However, once the infant is on extracorporeal support, the timing of surgical intervention is controversial. Some of the initial reports demonstrated a high complication rate associated with repair on ECMO, with bleeding being a significant issue.^{15,16} Subsequently, refinements in surgical technique such as the use of fibrin glue, diaphragmatic patching, aminocaproic acid, and abdominal wall patching have resulted in lower complication rates.^{17,18} Although there was a large range of timing for repair on ECMO, most patients underwent repair later (after 72 hours) in the ECMO course. The ideal timing for repair on ECMO is still, however, unclear.

A large number of the patients underwent patch repair of the diaphragm. This increased use of patch repair may be a result of more widespread repair on ECMO. Tissue edema and the desire to minimize bleeding by avoiding a dissection of the posterior rim of diaphragm make primary repair more difficult.¹⁷ Also there may be some benefit to attempting to recreate a dome shape to the diaphragm rather than a tight closure.¹⁹ A number of materials have been used to repair the defect, but PTFE is clearly the most widely used. An animal study demonstrated that PTFE made a good repair material when sewn to existing diaphragm.²⁰ Use of PTFE to reconstruct a diaphragm in the setting of complete agenesis may be problematic; however, because further growth could lead to chest wall deformity.²¹ Technical aspects of the repair have not previously been associated with changes in outcome. These data showed no difference in mortality rates of babies with right- or left-sided defects. Although the use of a patch was associated with a higher mortality rate, this is likely related to the patients' underlying disease severity. Demonstration of a significantly lower mortality rate when chest drains are not used is interesting

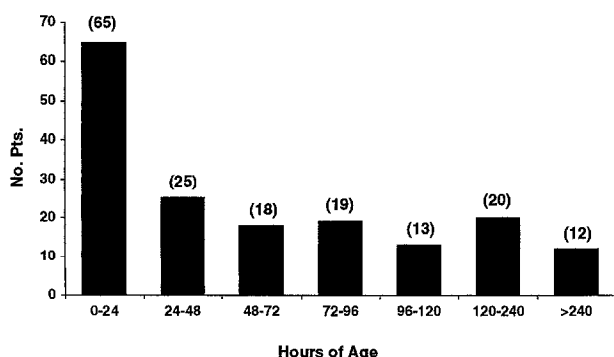


Fig 3. Age at surgery in patients who underwent CDH repair without ECMO. Numbers in parentheses represent the number of patients at each time period.

but cannot be overinterpreted because possible covariates have not been excluded. It certainly suggests that pleural drainage is not required in all patients undergoing repair of a CDH.

Optimal surgical management of CDH in infants would be best determined by prospective clinical trials. However, because CDH is so infrequently seen in most institutions, it will require a multicenter-based registry and study group to accomplish those trials. Although this is a preliminary report, several observations should

serve as departure points for future investigations. The mortality rate in patients without other anomalies and with small defects that can be repaired primarily is quite low. However, the high mortality rate in patients with large defects and other anomalies indicates that the CDH problem is far from solved. Future studies should focus on the standardized management of high-risk patients. Further multiinstitutional trials can be accomplished after this benchmark is established.

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Discussion

R.M. Arensman (Chicago, IL): I thank Dr Langham and Dr Lally for providing me with a copy of the manuscript, and I think you will all be very excited to see this, many of you of course having contributed to it.

As Dr Lally said, the great value of this is that primarily it gives us a snapshot of where we are and what we are currently doing in America to try and treat these children. As you can well believe, it raises far more questions than it answers.

We now of course have registries ongoing in biliary

atresia and trauma and extracorporeal life support and now we have the CDH. I think the great importance and significance of this registry has been recognized by your Executive Committee that yesterday decided to make modest monetary support of this registry because of its great value and what we hope is a powerful instrument to move us forward in this area.

In a 2-year period, in a very large number of patients, the survival rate is still only a little over 60%. There are some reasonably large institutional experiences now that

are reporting somewhere between 79% and 82% to 84% survival rates with the use of all the associated modalities: ECMO, nitric oxide, permissive hypercapnia, oscillatory ventilation, and surfactant therapy.

The great value, obviously, of a registry like this is our ability now to take this information and take these participating centers and to go forward in the next 2-year period to try and tease out a simple, easy question to answer, such as we have done through the POG and the CCSG studies over the years to move us forward in cancer therapy.

So I think I would conclude this presentation by asking Dr Langham and Dr Lally if they could identify one or two of the areas that we could all participate in the next 2 years to try and begin to answer the questions that

are obviously going to come out of the reading of this paper.

K.P. Lally (response): The more we learn about this anomaly and the more we look at the number of variables, the more it has become apparent that finding that question is pretty difficult.

Currently, we are trying to put together a protocol to look at steroid use for prenatal patients.

I think our first focus is to try and risk-stratify patients. It is very difficult to interpret the literature right now because everybody has different ways of stratifying these patients. The major focus that we have been working on for the last year or so is to develop at least an effective risk stratification system.

APPENDIX I

<i>Hospital</i>	<i>City</i>	<i>Country</i>
Arnold Palmer Hospital for Women and Children	Orlando, FL	
Boston Children's Hospital	Boston, MA	
Bowman Gray School of Medicine	Winston-Salem, NC	
Cardinal Glennon Children's Hospital	St Louis, MO	
Carolinas Medical Center	Charlotte, NC	
Children's Hospital of Akron	Akron, OH	
Children's Hospital of Alabama	Birmingham, AL	
Children's Hospital of Buffalo	Buffalo, NY	
Children's Hospital of Los Angeles	Los Angeles, CA	
Children's Hospital of Michigan	Detroit, MI	
Children's Hospital of Minneapolis	Minneapolis, MN	
Children's Hospital of Oakland	Oakland, CA	
Children's Hospital of Oklahoma	Oklahoma City, OK	
Children's Hospital of Wisconsin	Milwaukee, WI	
Children's Mercy Hospital	Kansas City, MO	
Children's National Medical Center	Washington, DC	
Cook Children's Hospital	Ft Worth, TX	
Duke University Medical Center	Durham, NC	
Egelston Children's Hospital	Atlanta, GA	
Hermann Children's Hospital	Houston, TX	
Hershey Medical Center	Hershey, PA	
James Whitcomb Riley Children's Hospital	Indianapolis, IN	
Kosair Children's Hospital	Louisville, KY	
Legacy Emanuel Children's Hospital	Portland, OR	
Loma Linda University Children's Hospital	Loma Linda, CA	
Lucile Salter Packard Children's Hospital	Palo Alto, CA	
Lutheran General Hospital	Park Ridge, IL	
Massachusetts General Hospital	Boston, MA	
Medical College of Georgia	Augusta, GA	
Medical University of South Carolina	Charleston, SC	
Miami Valley Hospital	Dayton, OH	
Ochsner Medical Center	New Orleans, LA	
Phoenix Children's Hospital	Phoenix, AZ	
Rainbow Babies and Children Hospital	Cleveland, OH	
Rockford Memorial Children's Hospital	Rockford, IL	
San Diego Children's Hospital	San Diego, CA	
Shands Children's Hospital/University of Florida	Gainesville, FL	
St Christopher's Children's Hospital	Philadelphia, PA	
St Francis Children's Hospital	Tulsa, OK	
St Joseph's Hospital and Medical Center	Phoenix, AZ	
St Louis Children's Hospital	St Louis, MO	
SUNY Brooklyn	Brooklyn, NY	
TC Thompson Hospital	Chattanooga, TN	
Texas Children's Hospital	Houston, TX	
Tulane University Hospital	New Orleans, LA	
University of California San Diego	San Diego, CA	
University of Kentucky Medical Center	Lexington, KY	
University of Michigan Medical Center	Ann Arbor, MI	
University of Nebraska Medical Center	Omaha, NE	
University of New Mexico Children's Hospital	Albuquerque, NM	
University of Virginia	Charlottesville, VA	
University of Texas Medical Branch at Galveston	Galveston, TX	
UPR Medical Center	San Juan, PR	
Vanderbilt University Children's Hospital	Nashville, TN	
Wilford Hall USAF Medical Center	San Antonio, TX	
Yale University Children's Hospital	New Haven, CT	
Ospedali Riuniti Bergamo	Bergamo	Italy
Royal Alexandra Children's Hospital	Edmonton, Alberta	Canada
Royal Children's Hospital	Parkville, Victoria	Australia
Royal Hospital for Sick Children	Glasgow, Yorkhill	Scotland
Salesi Children's Hospital	Ancona	Italy
Sydney Children's Hospital	Randwick, New South Wales	Australia